

bioclients

29 services, one shape

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Four apps, four gnomAD clients

4

APPS

3

GNOMAD CLIENTS THAT DISAGREED

2

MYGENE CLIENTS, ONE WITHOUT BATCH

chunk size

25, 20, or
unbounded

a missing gene

NULL, NA row, or
dropped

service down

crash, NULL, or
retry forever

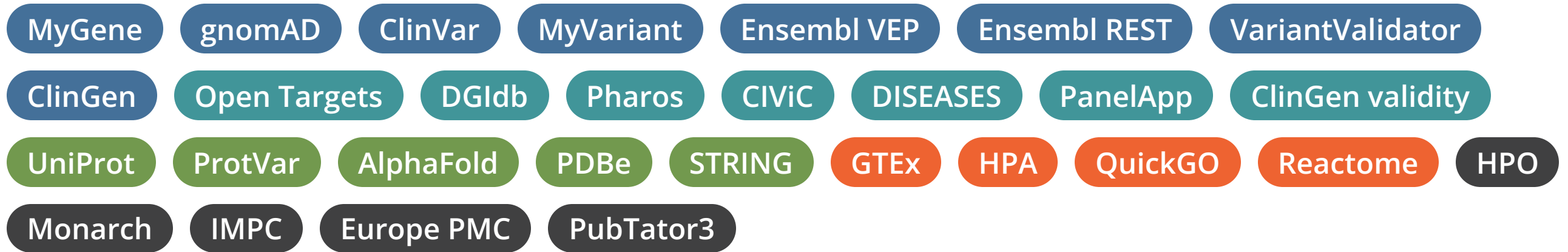
the trap

TTN resolved to TTR
in two of them

Same service, same author, three answers. That is not bad luck, that is a missing layer.



29 services



29

SERVICES

134

EXPORTED FUNCTIONS

57

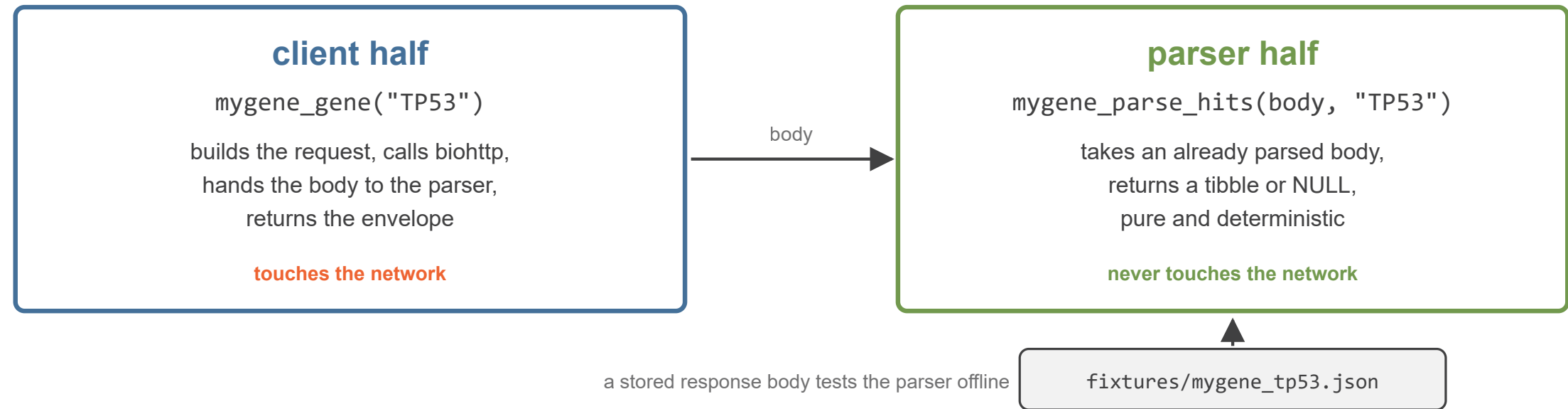
STORED FIXTURES

0

SHINY DEPENDENCIES

Genes, variants, disease, protein, expression, phenotype, literature. One envelope for all of it.

Client half, parser half



The parser is pure. The test suite runs without a network, and so can yours.

Nothing found is an answer

```
mygene_gene("NOTAREALGENE123")$status  
#> [1] "no_data"
```

```
mygene_gene("TP53")$status  
#> [1] "ok"
```

no_data

The service was reached. It has nothing for this query. Show “not found”.

error, timeout

The service was not reached, or failed. Show “try again”, not “not found”.

skipped

We did not ask. The host is paused. Say so.

A UI that says “not found” during an outage is lying to the user.

One row per input, in input order

```
mygene_genes(c("TP53", "NOTAREALGENE123", "BRAF"))$data
#>      symbol      name      entrez  ensembl_gene
#>      TP53      tumor protein p53      7157  ENSG00000141510
#>      NOTAREALGENE123  NA      NA      NA
#>      BRAF      B-Raf proto-oncogene      673  ENSG00000157764
```

the contract

N inputs, N rows, same order. A miss is an NA row in place.

what it prevents

A two row table shifts BRAF onto the second input. Every join after that is wrong.

Every batch function in the package keeps this. Zip results onto your input by position.

Traps, written down

TTN vs TTR

MyGene scores the alias higher. Exact symbol match wins.

HGNC is upper case

Every other MyGene field is lower case. Asking for `hgnc` returns nothing.

25 aliases

gnomAD's cost cap. Chunk at 20. Re-verified 2026-08-30.

Reactome is a bare array

A parser reaching for a `results` key reports every gene as pathway free.

```
# from R/mygene.R
# "This is the single most important thing in this file."
mygene_pick_hit(hits, "TTN")$entrezgene
#> [1] "7273"
```

Each trap has a comment with a date, and a test that pins it.

Limits are hard coded where they were verified

```
vep_default_throttle()  
#> $capacity 1    $fill_time_s 1  
  
bioclients:::clinvar_throttle()  
#> $capacity 3    $fill_time_s 1          # 10 with NCBI_API_KEY  
  
clinvar_classification(term)  
# throttle = clinvar_throttle(), secret_query = list(api_key = NCBI_API_KEY)
```

throttle

the service's documented rate, as the default

secret_query

→ the key attached at dispatch only

never in the cache key

→ never in a printed request

A caller who has not read the service docs still behaves.

Chaining sources

```
res <- mygene_gene("TP53")
uniprot_diseases(res$data$uniprot)$data
#>   name                acronym mim      causal
#>   Li-Fraumeni syndrome    LFS    151623  TRUE
#>   ...

opentargets_gene_diseases(res$data$ensembl_gene, size = 3)
```

Open Targets takes an Ensembl id

Resolve the symbol first. The client will not do it for you.

on purpose

A client that silently calls a second service makes a MyGene outage look like an Open Targets outage.

causal

“caused by variants” versus “may be involved”. Both arrive as disease comments.

One call, one service, one envelope. Composition happens in your code, where you can see it.

Start here

not on CRAN yet

```
pak::pak("samuelbharti/bioclients/pkg-r")
```

```
install.packages("bioclients",  
  repos = "https://samuelbharti.r-universe.dev")
```

The `/pkg-r` suffix is mandatory.

Try it in a minute

```
library(bioclients)  
res <- mygene_genes(c("TP53", "BRAF", "MLL"))  
res$status  
res$data
```

Docs: samuelbharti.com/bioclients